

PreACT 8/9 Printable Practice Exam 06

Math • Exam 02

35 questions • 40 minutes

PART A - QUESTIONS

M2-001

What is the value of $|-9| - \sqrt{16}$?

- A) -13
- B) -5
- C) 5
- D) 9
- E) 13

M2-002

What value of x satisfies $4x + 9 = 29$?

- A) 4
- B) 5
- C) 7
- D) 8
- E) 10

M2-003

Two angles of a triangle measure 47° and 68° . What is the measure of the third angle?

- A) 55°
- B) 65°
- C) 75°
- D) 111°
- E) 115°

M2-004

A plant is 14 centimeters tall when it is first measured. Its height increases at a constant rate, as shown in the table.

w (weeks)	0	3	6	9
h (centimeters)	14	20	26	32

Which equation gives the plant's height h after w weeks?

- A) $h = 2w + 14$
- B) $h = 3w + 14$
- C) $h = 6w + 14$
- D) $h = 14w + 2$
- E) $h = 14w + 6$

M2-005

What is the mean of the data set 6, 8, 8, 10, 13?

- A) 8
- B) 9
- C) 9.5

- D) 10
- E) 11

M2-006

What is $(6 \times 10^7) \div (3 \times 10^3)$ written in scientific notation?

- A) 2×10^3
- B) 2×10^4
- C) 2×10^{10}
- D) 3×10^4
- E) 3×10^{10}

M2-007

A restaurant bill is \$48. A customer leaves a tip equal to 18% of the bill. What is the total amount of the bill and tip?

- A) \$48.18
- B) \$54.40
- C) \$56.64
- D) \$57.60
- E) \$66.00

M2-008

On a map, 1 centimeter represents 12 kilometers. Two towns are 7.5 centimeters apart on the map. What is the actual distance between the towns?

- A) 19.5 kilometers
- B) 62.5 kilometers
- C) 72 kilometers
- D) 90 kilometers
- E) 120 kilometers

M2-009

A circular fountain has a radius of 7 meters. What is the circumference of the fountain?

- A) 7π meters
- B) 14π meters
- C) 28π meters
- D) 49π meters
- E) 98π meters

M2-010

A bag contains 5 red tiles, 3 blue tiles, and 2 green tiles. One tile is selected at random. What is the probability that the tile is not blue?

- A) $1/5$
- B) $3/10$
- C) $1/2$
- D) $7/10$
- E) $4/5$

M2-011

What is the value of $\sqrt{144} - \sqrt[3]{125}$?

- A) 5
- B) 7

- C) 9
- D) 12
- E) 17

M2-012

The first term of a sequence is 3. Each term after the first is twice the preceding term. What is the sixth term?

- A) 24
- B) 48
- C) 64
- D) 96
- E) 192

M2-013

A community group wants to collect at least \$500. It has already collected \$128 and expects to collect \$31 each day. What is the least number of additional days needed to reach the goal?

- A) 10
- B) 11
- C) 12
- D) 13
- E) 16

M2-014

Point R has coordinates $(-4, 2)$. It is translated 3 units to the right and 5 units down. What are the coordinates of the image of R?

- A) $(-9, 5)$
- B) $(-7, 7)$
- C) $(-1, -3)$
- D) $(1, -2)$
- E) $(1, 7)$

M2-015

The table shows whether 120 students have a library card.

	Has a card	Does not have a card	Total
Grade 8	36	24	60
Grade 9	42	18	60
Total	78	42	120

One student is selected at random from the students who do not have a library card. What is the probability that the selected student is in Grade 9?

- A) $\frac{3}{20}$
- B) $\frac{2}{7}$
- C) $\frac{3}{7}$
- D) $\frac{7}{20}$
- E) $\frac{7}{10}$

M2-016

A car travels 156 miles using 6 gallons of fuel at a constant rate. At this rate, how far can the car travel using 9.5 gallons of fuel?

- A) 171 miles
- B) 228 miles
- C) 234 miles

- D) 247 miles
- E) 296.4 miles

M2-017

A shopper buys 18 pieces of fruit consisting only of apples and pears. Apples cost \$2 each, pears cost \$3 each, and the total cost is \$44. How many pears does the shopper buy?

- A) 6
- B) 8
- C) 10
- D) 12
- E) 14

M2-018

A new video has 125 views when its view count is first recorded. The total number of views doubles each day. How many views will the video have 4 days after the first count?

- A) 500
- B) 1,000
- C) 1,500
- D) 2,000
- E) 4,000

M2-019

A cone has a radius of 4 centimeters and a height of 9 centimeters. What is its volume?

- A) 12π cubic centimeters
- B) 36π cubic centimeters
- C) 48π cubic centimeters
- D) 108π cubic centimeters
- E) 144π cubic centimeters

M2-020

A company wants to estimate the average commuting time of all its employees. Which method is most likely to produce a representative sample?

- A) Survey the 80 employees who arrive earliest.
- B) Survey all employees who ride the bus.
- C) Randomly select 80 employees from the complete employee roster.
- D) Ask for volunteers in the company's parking garage.
- E) Survey the 80 employees who live closest to the office.

M2-021

What is the value of $5^6 \div (5^2 \times 5)$?

- A) 5
- B) 25
- C) 125
- D) 625
- E) 15,625

M2-022

The equation $x^2 - 3x - 10 = 0$ has two solutions. What is the greater solution?

- A) -5
- B) -2

- C) 2
- D) 3
- E) 5

M2-023

A gym charges a monthly fee of \$24 plus \$8 for each fitness class attended. One member's total charge for a month is \$72. How many fitness classes did the member attend?

- A) 4
- B) 5
- C) 6
- D) 8
- E) 9

M2-024

Two triangles are similar. A 10-centimeter side of the smaller triangle corresponds to a 15-centimeter side of the larger triangle. If another side of the smaller triangle is 11 centimeters, what is the length of its corresponding side in the larger triangle?

- A) 13.5 centimeters
- B) 15.5 centimeters
- C) 16.5 centimeters
- D) 21 centimeters
- E) 26 centimeters

M2-025

Company A charges $y = 4x + 18$ dollars to rent a bicycle for x hours. Company B charges a starting fee plus a constant amount per hour, as shown in the table.

x (hours)	0	2	4
y (dollars)	10	20	30

For a 6-hour rental, how does Company A's charge compare with Company B's charge?

- A) Company A charges \$2 less.
- B) Company A charges \$2 more.
- C) Company A charges \$8 less.
- D) Company A charges \$8 more.
- E) The two companies charge the same amount.

M2-026

A rectangular room is 15 feet long and 12 feet wide. A contractor plans to buy 10% more flooring than the room's area to allow for waste. Flooring is sold only in whole boxes, and each box covers 20 square feet. What is the least number of boxes the contractor must buy?

- A) 9
- B) 10
- C) 11
- D) 18
- E) 20

M2-027

What value of x satisfies $4(2x - 3) = 3x + 23$?

- A) 5
- B) 6
- C) 7

- D) 8
- E) 11

M2-028

Which of the following is an irrational number between 4 and 5?

- A) $\sqrt{15}$
- B) 4.25
- C) $\sqrt{20}$
- D) $\sqrt{25}$
- E) $\sqrt[3]{125}$

M2-029

What is the midpoint of the line segment with endpoints $(-3, 5)$ and $(7, -1)$?

- A) $(-5, 3)$
- B) $(1, 2)$
- C) $(2, 2)$
- D) $(2, 3)$
- E) $(5, 4)$

M2-030

A linear function f passes through the points $(-1, 4)$ and $(3, 12)$. What is $f(5)$?

- A) 10
- B) 12
- C) 14
- D) 16
- E) 18

M2-031

One integer is selected at random from the integers 1 through 24. What is the probability that the selected integer is a multiple of 3 or a multiple of 5?

- A) $\frac{1}{3}$
- B) $\frac{3}{8}$
- C) $\frac{5}{12}$
- D) $\frac{11}{24}$
- E) $\frac{1}{2}$

M2-032

A water tank holds 2,400 liters when full and is currently 35% full. Water flows into the tank at 18 liters per minute. How many minutes will it take for the tank to become 80% full?

- A) 40 minutes
- B) 47 minutes
- C) 60 minutes
- D) 80 minutes
- E) 107 minutes

M2-033

A theater sells 120 tickets consisting only of adult and youth tickets. Adult tickets cost \$14, youth tickets cost \$9, and the total revenue is \$1,430. How many youth tickets are sold?

- A) 40
- B) 50

- C) 60
- D) 70
- E) 80

M2-034

A solid is made by joining 2 identical cubes, each with an edge length of 4 centimeters, along one complete face. What is the surface area of the resulting solid?

- A) 96 square centimeters
- B) 128 square centimeters
- C) 144 square centimeters
- D) 160 square centimeters
- E) 192 square centimeters

M2-035

A student has 5 quiz scores: 72, 78, 82, 88, and 90. The score of 72 is replaced by a score of 84. Which statement correctly describes how the mean and range change?

- A) The mean increases by 2.4 points, and the range decreases by 6 points.
- B) The mean increases by 6 points, and the range decreases by 2.4 points.
- C) The mean increases by 12 points, and the range decreases by 6 points.
- D) The mean increases by 2.4 points, and the range increases by 6 points.
- E) The mean and range each increase by 2.4 points.

PART B - ANSWER KEY + EXPLANATIONS

M2-001 - Correct: C) 5

Explanation: $|-9| = 9$ and $\sqrt{16} = 4$, so $|-9| - \sqrt{16} = 9 - 4 = 5$.

M2-002 - Correct: B) 5

Explanation: Subtract 9 from both sides to get $4x = 20$, and then divide by 4 to get $x = 5$.

M2-003 - Correct: B) 65°

Explanation: The angles of a triangle sum to 180° , so the third angle is $180^\circ - 47^\circ - 68^\circ = 65^\circ$.

M2-004 - Correct: A) $h = 2w + 14$

Explanation: The height is 14 centimeters at week 0 and increases by 6 centimeters every 3 weeks, a rate of 2 centimeters per week. Therefore, $h = 2w + 14$.

M2-005 - Correct: B) 9

Explanation: The sum of the 5 values is 45, so the mean is $45 \div 5 = 9$.

M2-006 - Correct: B) 2×10^4

Explanation: Divide the coefficients to get 2, and divide the powers of 10: $10^7 \div 10^3 = 10^4$. Therefore, the result is 2×10^4 .

M2-007 - Correct: C) \$56.64

Explanation: The tip is $\$48 \times 0.18 = \8.64 . Adding it to the bill gives $\$48 + \$8.64 = \$56.64$.

M2-008 - Correct: D) 90 kilometers

Explanation: Each centimeter represents 12 kilometers, so $7.5 \times 12 = 90$ kilometers.

M2-009 - Correct: B) 14π meters

Explanation: The circumference is $2\pi r$. With $r = 7$, it is $2 \times \pi \times 7 = 14\pi$ meters.

M2-010 - Correct: D) $7/10$

Explanation: There are 10 tiles total and 7 are not blue, so the probability is $7/10$.

M2-011 - Correct: B) 7

Explanation: $\sqrt{144} = 12$ and $\sqrt[3]{125} = 5$, so $\sqrt{144} - \sqrt[3]{125} = 12 - 5 = 7$.

M2-012 - Correct: D) 96

Explanation: Starting with 3 and doubling 5 times gives 3, 6, 12, 24, 48, 96. Therefore, the sixth term is 96.

M2-013 - Correct: C) 12

Explanation: The group still needs $\$500 - \$128 = \$372$. Since $\$372 \div \$31 = 12$, it needs at least 12 additional days.

M2-014 - Correct: C) $(-1, -3)$

Explanation: Moving $(-4, 2)$ right 3 units changes x to -1 , and moving it down 5 units changes y to -3 .

M2-015 - Correct: C) $3/7$

Explanation: Of the 42 students without a library card, 18 are in Grade 9. The probability is $18/42 = 3/7$.

M2-016 - Correct: D) 247 miles

Explanation: The car travels $156 \div 6 = 26$ miles per gallon. With 9.5 gallons, it travels $26 \times 9.5 = 247$ miles.

M2-017 - Correct: B) 8

Explanation: Let a be apples and p be pears. Then $a + p = 18$ and $2a + 3p = 44$. Substituting $a = 18 - p$ gives $36 + p = 44$, so $p = 8$.

M2-018 - Correct: D) 2,000

Explanation: Four days produce 4 doublings, so the number of views is $125 \times 2^4 = 125 \times 16 = 2,000$.

M2-019 - Correct: C) 48π cubic centimeters

Explanation: A cone's volume is $(1/3)\pi r^2 h$. Substituting $r = 4$ and $h = 9$ gives $(1/3) \times \pi \times 4^2 \times 9 = 48\pi$ cubic centimeters.

M2-020 - Correct: C) Randomly select 80 employees from the complete employee roster.

Explanation: A random selection from the full roster gives every employee a chance to be included and avoids choosing a group based on commuting habits.

M2-021 - Correct: C) 125

Explanation: Since $5^2 \times 5 = 5^3$, the expression is $5^6 \div 5^3 = 5^3 = 125$.

M2-022 - Correct: E) 5

Explanation: Factoring gives $x^2 - 3x - 10 = (x - 5)(x + 2)$, so the solutions are 5 and -2 . The greater solution is 5.

M2-023 - Correct: C) 6

Explanation: Subtract the \$24 monthly fee from \$72 to get \$48. At \$8 per class, the member attended $\$48 \div \$8 = 6$ classes.

M2-024 - Correct: C) 16.5 centimeters

Explanation: The scale factor is $15/10 = 1.5$. Therefore, the corresponding side is $11 \times 1.5 = 16.5$ centimeters.

M2-025 - Correct: B) Company A charges \$2 more.

Explanation: For 6 hours, Company A charges $4(6) + 18 = \$42$. Company B's table has a rate of \$5 per hour and a starting charge of \$10, so it charges $5(6) + 10 = \$40$. Company A charges \$2 more.

M2-026 - Correct: B) 10

Explanation: The room's area is $15 \times 12 = 180$ square feet. Including 10% extra requires $180 \times 1.10 = 198$ square feet. Since $198 \div 20 = 9.9$, the contractor must buy 10 whole boxes.

M2-027 - Correct: C) 7

Explanation: Expanding gives $8x - 12 = 3x + 23$. Then $5x = 35$, so $x = 7$.

M2-028 - Correct: C) $\sqrt{20}$

Explanation: Since $16 < 20 < 25$, $4 < \sqrt{20} < 5$. Because 20 is not a perfect square, $\sqrt{20}$ is irrational.

M2-029 - Correct: C) (2, 2)

Explanation: Average the x-coordinates and y-coordinates: $((-3 + 7)/2, (5 + (-1))/2) = (2, 2)$.

M2-030 - Correct: D) 16

Explanation: The slope is $(12 - 4)/(3 - (-1)) = 8/4 = 2$. The function is $f(x) = 2x + 6$, so $f(5) = 16$.

M2-031 - Correct: D) 11/24

Explanation: From 1 through 24 there are 8 multiples of 3 and 4 multiples of 5. The number 15 is counted in both groups, so there are $8 + 4 - 1 = 11$ favorable integers. The probability is 11/24.

M2-032 - Correct: C) 60 minutes

Explanation: The tank must increase from 35% to 80%, a change of 45% of 2,400 liters. That is $0.45 \times 2,400 = 1,080$ liters. At 18 liters per minute, the time is $1,080 \div 18 = 60$ minutes.

M2-033 - Correct: B) 50

Explanation: If all 120 tickets were youth tickets, the revenue would be \$1,080. Each adult ticket adds \$5, so the number of adult tickets is $(\$1,430 - \$1,080) \div \$5 = 70$. Therefore, $120 - 70 = 50$ youth tickets were sold.

M2-034 - Correct: D) 160 square centimeters

Explanation: Two separate cubes have $2 \times 6 = 12$ square faces. Joining them hides 2 faces, leaving 10 exposed faces. Each face has area $4^2 = 16$ square centimeters, so the surface area is $10 \times 16 = 160$ square centimeters.

M2-035 - Correct: A) The mean increases by 2.4 points, and the range decreases by 6 points.

Explanation: Replacing 72 with 84 increases the total by 12, so the mean increases by $12 \div 5 = 2.4$. The original range is $90 - 72 = 18$, and the new ordered scores are 78, 82, 84, 88, 90, with range $90 - 78 = 12$, a decrease of 6.